

INTERNATIONAL GCSE PHYSICS

9203/1

Paper 1

Mark scheme

November 2023

Version: 1.0 Final



2 3 B Y 9 2 0 3 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	an energy resource that can be replaced as quickly as it is used		1	AO1 3.2.3d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	biofuel		1	AO1 3.2.3d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	coal		1	AO3 3.2.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	mass of carbon dioxide per 1 kW h = 220 (g)		1	AO2 3.2.3c
	mass = 220×40		1	
	mass = 8 800 (g)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	$E = 2.0 \times 6.0$		1	AO2 3.6.5f
	$E = 12$ (kW h)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	cost = 12×0.35	allow ecf from question 01.5	1	AO2 3.6.5f
	cost = \$ 4.20		1	

Total Question 1		10
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	a quantity that has magnitude and a direction		1	AO1 3.1.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	momentum velocity		1 1	AO1 3.1.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	C		1	AO1 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	A		1	AO1 3.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	$171\,500 = m \times 9.8 \times 250$ $m = \frac{171\,500}{9.8 \times 250}$ $m = 70 \text{ (kg)}$		1 1 1	AO2 3.2.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	$1.25 = \frac{400}{t}$		1	AO2 3.1.2c
	$t = \frac{400}{1.25}$		1	
	$t = 320 \text{ (s)}$		1	

Total Question 2		11
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	any two from: - distance of wax from holder - length / diameter of metal bar - mass of wax - position of Bunsen burner	allow distance of wax from Bunsen burner	2	AO4 3.4.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	iron smallest spread about the mean	allow the results have the smallest range	1 1	AO3 3.4.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	uncertainty = $(\pm) 3$ (s)	allow 3.0 (s) allow one mark for range = $68 - 62 = 6$	2	AO2 3.4.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(at the heated end of the metal bar) ions vibrate more	allow atoms for ions throughout	1	AO1 3.4.2a
	and the free / conduction electrons move faster (through the metal)		1	
	electrons and ions collide with neighbouring ions transferring energy along the bar		1	

Total Question 3		9
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	3.0 V		1	AO1 3.5.1p

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2 mark with 04.3	0.25 A		1	AO1 3.5.1r

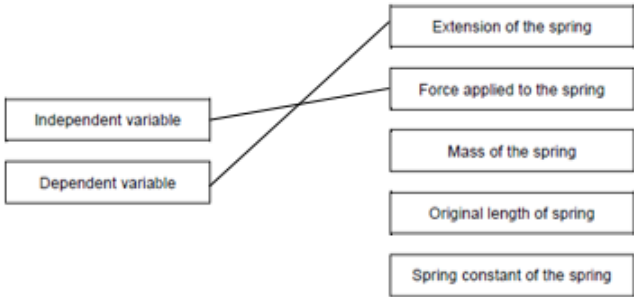
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3 mark with 04.2	$V = 0.25 \times 5.0$	allow ecf from Question 04.2	1	AO2 3.5.1h 4-5
	$V = 1.25 \text{ (V)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	$0.50 = \frac{Q}{22}$		1	1 × AO1 3 × AO2 3.5.1c
	$Q = 0.50 \times 22$		1	
	$Q = 11$		1	
	coulombs		1	
		allow C		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	less than 12.0 Ω		1	AO3 3.5.1s

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	the resistance is constant (at constant temperature)		1	AO1 3.5.1i
	so current is directly proportional to potential difference		1	
	therefore the graph is a straight line through the origin		1	

Total Question 4		12
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Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	independent variable - force applied to the spring dependent variable - extension of the spring  <pre> graph LR IV[Independent variable] --- F[Force applied to the spring] DV[Dependent variable] --- E[Extension of the spring] </pre> do not accept more than one line drawn from a box on the left.		1 1	AO4 3.1.1h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	any two pairs from: • spring going into the eye • wear safety glasses • hair gets caught in the spring • tie hair up • slotted masses fall off the spring • place foam under slotted masses • stand falls off bench • clamp the stand to the bench	allow move stand to the centre of the bench	4	AO4 3.1.1h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	gradient = $\frac{6-0}{0.12-0}$	allow use of $F=ke$ and any correct pair of readings (taken from the line of best fit)	1	AO2 3.1.1h
	$k = 50 \text{ (N/m)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	(for the runner with the greater mass) the force on the spring is increased	dependent on mp1	1	AO3 3.1.1h
	therefore a spring with a greater spring constant is needed		1	
	to produce the same compression		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	time for foot to stop once the shoe touches the ground increases		1	AO3 3.1.4c
	so rate of change of momentum decreases		1	
	(force is reduced because) force is equal to rate of change of momentum		1	

Total Question 5		14
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	vertical line through X	allow line / arrow drawn vertically downwards from X	1	AO1 3.1.8c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	the point at which the mass / weight of the object can be considered to act		1	AO1 3.1.7a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	widen the base	allow sensible suggestion that widens the base	1	AO3 3.1.8c
	lower the centre of mass	allow sensible suggestion that lowers the centre of mass	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	Level 2: Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.		3–4	AO1 3.1.7a
	Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.		1–2	
	No relevant content		0	
	Indicative content • make a hole near the edge of the shape • place the pin through the hole • hold a pin in the clamp on the retort stand, suspend shape from the pin so it swings freely • hang a plumbline from the pin • draw a line on the shape along the plumbline • repeat for other holes at different positions on the shape • where the drawn lines cross is the centre of mass	allow description of a plumbline		

Total Question 6		8
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	light rays are refracted	allow change direction	1	AO1 3.3.1e
	the light waves slow down	allow the glass has a different optical density to the surrounding air	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	convex	allow converging	1	AO1 3.3.6b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	upright		1	AO1 3.3.6b
	virtual		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	$2.5 = \frac{24}{\text{object height}}$		1	AO2 3.3.6g
	$\text{object height} = \frac{24}{2.5}$		1	
	object height = 9.6 (mm)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	the angle of incidence must be greater than the critical angle		1	AO1 3.3.5g
	light must be travelling from a more optically dense medium into a less optically dense medium	mp2 maybe awarded for an answer in terms of refractive indices	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.6	refractive index = $\frac{\sin 35^\circ}{\sin 24^\circ}$		1	AO2 3.3.5ef
	refractive index = 1.41		1	
	$1.41 = \frac{1}{\sin c}$	for subsequent marks to be awarded the correct equation must have been used to calculate the refractive index		
	$\sin c = \frac{1}{1.41}$	allow correct substitution using incorrectly calculated refractive index	1	
	$c = 45 (^\circ)$	allow correct rearrangement using incorrectly calculated refractive index	1	
		allow 45.2...	1	

Total Question 7		15
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Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	38 protons	either order	1	AO1 3.7.1f
	51 neutrons		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	neutron (in the nucleus) decays into a proton		1	AO1 3.7.2e
	and an electron (which is emitted as a beta particle)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	${}_{38}^{89}\text{Sr} \longrightarrow {}_{39}^{89}\text{Y} + {}_{-1}^0\beta$	1 mark for the top row 89, 0 1 mark for the bottom row 39, -1	1 1	AO2 3.7.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	$\frac{202}{50.5} = 4 \text{ half lives}$		1	AO2 3.7.2h
	$192 \times \left(\frac{1}{2}\right)^4$		1	
	or			
	$192 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$			
	count rate = 12 (counts per second)	allow 192, 96, 48, 24, 12	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	the ionising radiation kills the cancer cells		1	AO3 3.7.2i 6-7 8-9
	which outweighs the risk of damaging healthy cells		1	

Total Question 8		11
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